

BORONG XU

Research Assistant at SLAI

+852 8491 8005

borong180-github-io.vercel.app/

borong180

borongxu@outlook.com

borong-xu-52829a293

ABOUT ME

Research Assistant at the Shenzhen Loop Area Institute (SLAI) and a graduate of HKUST's Computer Engineering (CEG) program, ranked 1/98 by major GPA. Passionate about robotics and embodied intelligence, with research experience spanning visual-inertial odometry, SLAM, multi-camera localization, and quadruped control, as well as deep learning, vision transformers, and LLM-based agent systems. Eager to push the boundaries of autonomous systems in real-world deployments.

EDUCATION

The Hong Kong University of Science and Technology (HKUST)

BACHELOR OF ENGINEERING IN COMPUTER ENGINEERING, FIRST CLASS HONORS

Hong Kong SAR

Graduated Jun 2026

- GPA 3.84/4.3
- Major GPA 4.03/4.3 (Ranked 1/98 in program)

University of New South Wales (UNSW Sydney)

VISITING STUDENT (EXCHANGE), COMPUTER SCIENCE AND ENGINEERING

Sydney, Australia

Feb 2025 - May 2025

High School Affiliated to Renmin University of China (RDFZ)

HIGH SCHOOL DIPLOMA(GAOKAO)

Beijing, China

Graduated Jun 2022

- Activities:
 - FIRST Robotics Club, Chief Engineer (season 2021)
 - FIRST Robotics Club, Alumni Mentor (season 2022-2024)

EXPERIENCE

Shenzhen Loop Area Institute (SLAI)

RESEARCH ASSISTANT

Shenzhen, China

Jun 2026 - Present

- Research on robotics and embodied intelligence.

PROJECT EXPERIENCE

LEGO Assembly Understanding

ONGOING RESEARCH PROJECT, JOINTLY SUPERVISED BY HKUST RFL AND SLAI

Supervised by Prof. Ziqi Wang

Mar 2026 - Present

- Research on the automatic understanding of LEGO assembly, combining computer vision and planning to recover step-by-step assembly structure from visual instructions and 3D models.

ELEC5660 Introduction to Aerial Robotics

COURSE PROJECT, HKUST

Supervised by Prof. Shaojie Shen

Feb 2026 - May 2026

- Developed aerial robot algorithms from scratch, including visual-inertial odometry, Extended Kalman Filter (EKF) state estimation, and autonomous path planning within a simulation environment, enabling robust UAV localization, mapping, and navigation.

Quadruped Robot Control

RESEARCH PRACTICUM, UNSW SYDNEY

Supervised by Prof. Maurice Pagnucco and Prof. Yang Song

Feb 2025 - Aug 2025

- Built and programmed a Unitree GO2 quadruped robot to perform complex tasks, integrating depth camera for accurate object localization (localization error < 10cm) and LiDAR for SLAM.

Underwater Localization System

FINAL YEAR PROJECT, HKUST

Supervised by Prof. Huan Yin and Prof. Fumin Zhang

Jun 2025 - May 2026

- Developed a Multi-Camera Vision-Based Localization System for Tracking Multiple Autonomous Underwater Vehicles (AUVs). Reduced localization error in complex underwater lighting scenarios, improving positioning accuracy to within 10cm in a 1m×1m range.



SKILLS

- **C/C++:** Eigen, PCL, STL, STM32, ESP32
- **Python:** PyTorch, OpenCV, Numpy
- **ROS/ROS2:** Navigation2, Gazebo, Slam Toolbox, TF
- **Robotics:** SLAM, VIO, EKF, ICP, Path Planning, Control
- **Deep Learning:** YOLO, ViT, RNN, GAN, VAE, Diffusion, LLM
- **Machine Learning:** GLM, SVM, PCA, XGBoost, HMM
- **Hardware:** Circuit Design, Soldering, CAD, 3D Printing

LANGUAGES

English • Fluent (IELTS 7.5)
Mandarin • Mother tongue

HONORS

2022-2026	Dean's List , 5 times	HKUST
2023-2026	Continuing Undergraduate Students Scholarship , 10,000 HKD per year	HKUST
2023	Bronze Medal , The Innovation Competition: Our Livable City, We Engineer	HKIE
2022	Entry Scholarship , 70,000 HKD	HKUST